Performance of Wearable and Nearable Sensors in Detecting Motor Activity During Sleep

Olivia Lecomte

Faculty of Science and Medicine, UNIFR

Analysing Health Data from Wearable Biosensors

Dr. Oriella Gnarra

June 7th, 2024

Introduction

Polysomnography (PSG) was introduced in the 1960s to monitor sleep in laboratory environments (Perez-Pozuelo et al., 2020). This comprehensive method utilises several sensors to simultaneously measure (1) brain activity via electroencephalography (EEG), (2) airflow, (3) respiratory effort and rate, (4) blood oxygen levels, (5) body position, (6) eye movement, (7) muscular activity via electromyography (EMG), and (8) heart rate (Perez-Pozuelo et al., 2020). Despite its accuracy, PSG requires a specially trained clinician to set up and interpret the results (Chinoy et al., 2020; Perez-Pozuelo et al., 2020). This, combined with the fact that these tests are often performed in a laboratory setting, is expensive and impractical for longitudinal sleep studies (Perez-Pozuelo et al., 2020). Furthermore, the intrusive nature of the equipment can disrupt sleep and introduce bias into the results (Chinoy et al., 2020). Despite this, PSG remains the gold standard for sleep monitoring and serves as the ground truth for many comparison studies (Perez-Pozuelo et al., 2020). These challenges have motivated researchers and clinicians to search for alternative methods (Perez-Pozuelo et al., 2020).

One validated alternative is actigraphy, a method which tracks physical activity during sleep with the help of a wearable device fitted with an accelerometer (Chinoy et al., 2020). This method is less intrusive than PSG and can be used to monitor sleep in the wearer's natural environment (Chinoy et al., 2020). Movements detected by the accelerometer are leveraged to indirectly determine whether the wearer is awake or asleep (Chinoy et al., 2020). As a result, these devices tend to overestimate sleep and underestimate wakefulness time when wearers are awake but not moving (Chinoy et al., 2020).

Recently, clinicians have turned to the evaluation of consumer-grade fitness trackers as a viable alternative to PSG. Most fitness trackers are equipped with accelerometers similar to those used in actigraphy and some have been used for sleep research (Chinoy et al., 2020). Researchers are leveraging the growing public interest in wearable devices and the potential to complete long-term sleep studies in naturalistic environments (Chinoy et al., 2020).

Both wearable and nearable devices have been investigated as viable alternatives for PSG. For instance, the use of personalised and general machine learning algorithms to decode accelerometer data collected by the Oura (Altini et al., 2021) and Happy (Grandner et al., 2022) rings, achieved high accuracy in detecting sleep stages. The Xiaomi Mi Band 5, one of the most popular activity wristbands, is also equipped with an accelerometer (Concheiro-Moscoso et al., 2023). It demonstrated an accuracy of 78%, a sensitivity of 89% and a specificity of 35% as compared to PSG measures with better performance being attributed to wearers with no known sleep disorders (Concheiro-Moscoso et al., 2023).

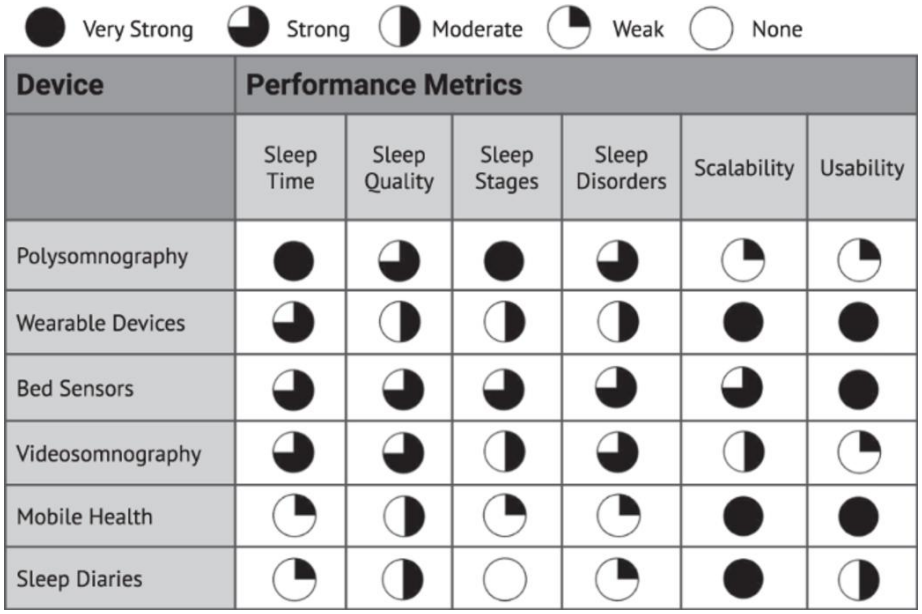
In a review of numerous wearable and nearable devices, Chinoy et al. (2020) reported that most devices could match or surpass the performance of traditional actigraphy methods in detecting wakefulness whereas both methods exhibited high accuracy for sleep detection. Notably, while actigraphy was significantly less accurate than PSG in detecting sleep and wakefulness stages, many consumer devices were not (Chinoy et al., 2020).

Another area of interest is the monitoring of periodic limb movements in sleep (PLMS), a neurological disorder characterized by repetitive limb movements that can cause sleep disturbances (Hornyak et al., 2006; Joseph & Nagalli, 2024; *Periodic Limb Movement Disorder (PLMD) and Restless Legs Syndrome (RLS) - Brain, Spinal Cord, and Nerve Disorders*, 2022). Though typically detected using PSG, other less invasive wearables such as accelerometer bands worn across the metatarsal (Kye et al., 2017) and electromyography-embedded socks (Eguchi et al., 2019), show promise for less invasive detection of PLMS.

In summary, while PSG serves as the benchmark for the accurate monitoring of PLMS, its limitations with regards to cost, practicality, and intrusiveness have fuelled the search for alternate options such as wearable and nearable devices. Although more prone to noise, wearable and nearable devices offer scalability and usability that cannot be matched by PSG (Figure 1) (Perez-Pozuelo et al., 2020).

Figure 1

Comparison of Various Wearable and Nearable Devices



Note. From Perez-Pozuelo, I., Zhai, B., Palotti, J., Mall, R., Aupetit, M., Garcia-Gomez, J. M., Taheri, S., Guan, Y., & Fernandez-Luque, L. (2020). The future of sleep health: A data-driven revolution in sleep science and medicine. *Npj Digital Medicine*, *3*(1), 1–15. https://doi.org/10.1038/s41746-020-0244-4

Method

This report focuses on the performance of two nearable bed sensor devices, the EMFIT 05 and 19, and a wearable wrist band, the Empatica E4, compared with the SOMNOmedics PSG system.

Participants

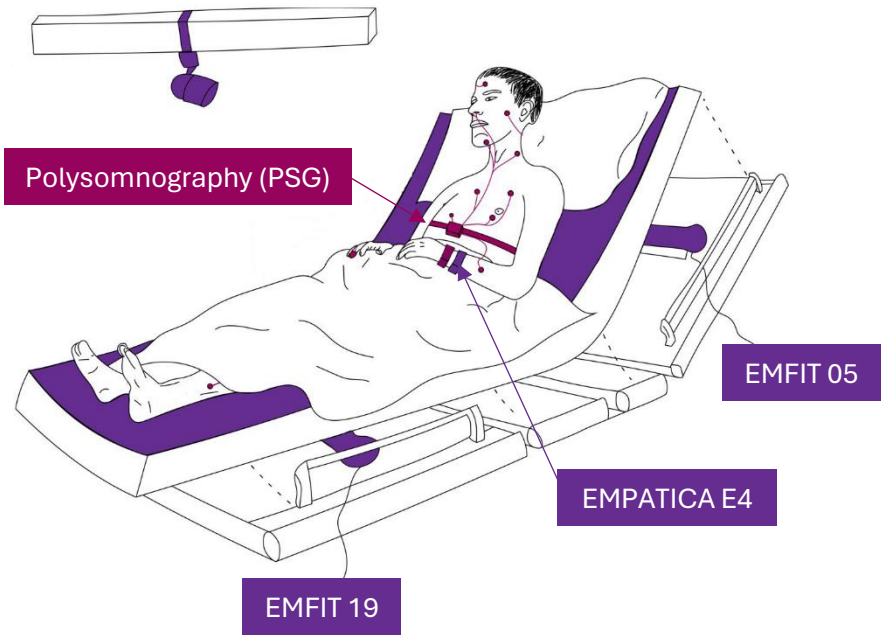
A total of 115 adults (62 females) aged between 18-85 (*M* = 47) participated in an overnight sleep study comparing the performance of various wearable and nearable devices. Of these participants, 68% had a known sleeping disorder. The analysis focused on the 71 participants with a functional PSG file.

Material

This study involved the comparison of two nearable and one wearable device against a PSG system. The SOMNOmedics PSG system is a wireless, compact, and robust solution for stationary sleep lab environments. The data collected from these sensors, acquired at a rate of 4Hz, serves as the ground truth for further comparisons. The EMFIT QS is a nearable device which consists of quazi-pietzoelectric sensors installed under the mattress. Through ballistocardiography data analysis, this device collects physiological data such as heart contractions, breathing, and body movements at a sampling rate of 0.25Hz. For this study, two EMFIT devices were fitted, the EMFIT 05 was located beneath the participant's upper torso, and the EMFIT 19 beneath their legs (Figure 2). Finally, the EMPATICA E4 device is a wearable wristband fitted with four sensors. Our analysis focused on the motion data collected by the 3-axis accelerometer at a sampling rate of 32Hz.

Figure 2

Studied Wearable and Nearable Devices



Analysis

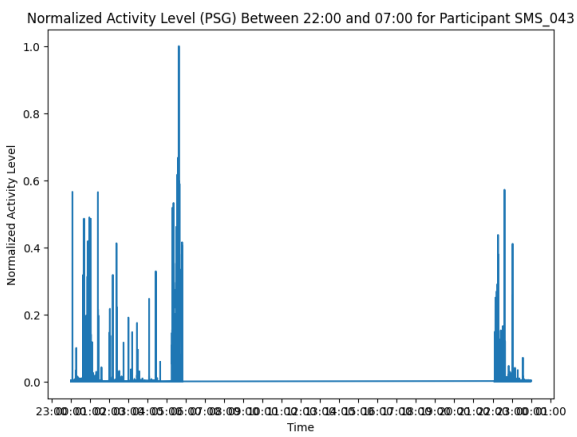
Data from one night of sleep was recorded using the PSG sleep system, two nearable devices, and one wearable device. The data was processed and analyed using Python 3.10. For the analyses of the PSG and nearable devices, data was loaded for each participant with a valid file folder and filtered to include only the hours between 22:00 and 07:00. Each nearable sensor's data was then normalized to a range between 0 and 1 using min-max normalization to facilitate inter-sensor comparisons (Equation 1).

$$normalised\ activity = \frac{activity - min(activity)}{max(activity) - min(activity)} \tag{1}$$

After pre-processing, each nearable sensor was compared to the PSG benchmark. The normalized PSG signal was visually inspected alongside the corresponding EMFIT 05 signal where available. This inspection revealed errors in the overnight time mapping of the PSG files for five participants, who were subsequently excluded from further analyses (Figure 3). For the 35 participants with both functional PSG and EMFIT 05 signals, cross-correlation was performed using the 'same' method, which outputs an array with the same dimensions as the input through single padding. The correlation at lag 0 was used to measure the correlation of the signals before alignment. The signals were then realigned based on the index of the lag with the highest correlation and cross-correlated again. For each of these 35 participants, plots were generated showing the PSG signal overlaid on the EMFIT 05 signal for the filtered period, as well as the cross-correlation before and after alignment. The realigned correlation was again measured at lag 0, and mean correlations were calculated before and after realignment. The analysis of the EMFIT 19 signals followed the same procedure, with a total of 37 participants included in the comparison.

Figure 3

Example of a Participant with Time Mapping Error



The pre-processing of the Empatica data followed a similar procedure, with data filtered to include the hours between 22:00 and 07:00. However, an additional normalization function was used to convert the 3D signal into a single scalar value representing the acceleration magnitude of the vector (Equation 2). This was applied before a second normalisation which bound the values between 0 and 1. A total of 56 participants were compared.

$$normalised\ activity = \sqrt{(x^2 + y^2 + z^2)} \tag{2}$$

Results

The analysis compared the signals recorded by EMFIT 05, EMFIT 19, and Empatica E4 against PSG. The findings, including the number of compared participants and the mean correlation scores before and after realignment, are summarised in the following table (Table 1).

Table 1

Summary of Results from Analysis of Wearable and Nearable Devices Compared to PSG

Metric	EMFIT 05	EMFIT 19	Empatica E4
Number of compared participants	35	37	56
Mean correlation before alignment	0.32	0.26	N/A
Mean correlation after alignment	0.97	0.99	0.58

Note. The reported correlations were calculated at lag 0 before and after alignment using cross-correlation with the 'same' method.

Discussion

Though the correlations between the tested nearable devices and PSG system are high, unresolved challenges related to the cross-correlation and signal realignment suggest that these results are not reliable. The primary issue affecting the cross-correlation accuracy stems from the differing sampling rates between the PSG system and the wearable and nearable devices. Various methods were attempted to mitigate this issue.

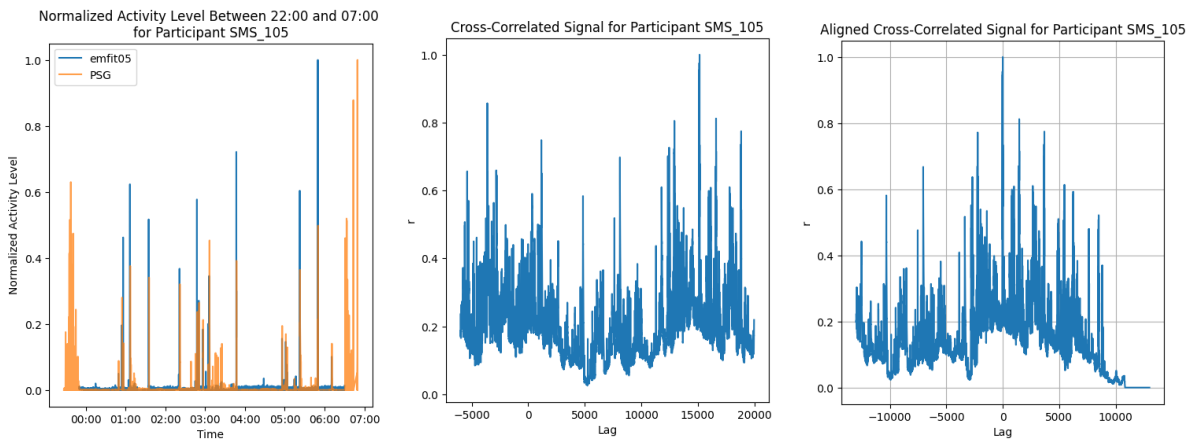
According to the device documentation, the PSG system samples 16 times more frequently than the EMFIT devices. The initial down sampling attempt aimed to index samples at equally spaced intervals to create two representative samples of equal length. However, the PSG signal was not always the longer signal, indicating probable data loss or recording inconsistencies. A more sophisticated approach involved matching samples based on the most similar time points between the two arrays. Despite its potential, this method required excessive processing time and did not complete within a reasonable timeframe. Consequently, the accuracy of the correlations cannot be trusted due to the non-identical sampling rates between

the two arrays. These problems are further compounded by the realignment process which used zero padding to ensure the arrays were of identical lengths.

A visual inspection of the resulting plots confirms that the high correlations are unreliable. While most correlations between the PSG system and nearable devices frequently reached 1, the overlaid signals plots clearly suggest otherwise. Firstly, the recording of PSG signals often begins before and ends after the range of signals captured by the wearable and nearable devices (Figure 4 & 5), which should prevent from attaining a perfect correlation. Secondly, while a visual inspection of the cross-correlation plots suggests a correct realignment of the signals based on index, some plots exhibit a discrepancy between the plotted correlation at lag 0 and the printed correlation, despite originating from the same data (Figure 4).

Figure 4

Comparison Between PSG system and EMFIT 05 for Participant 105



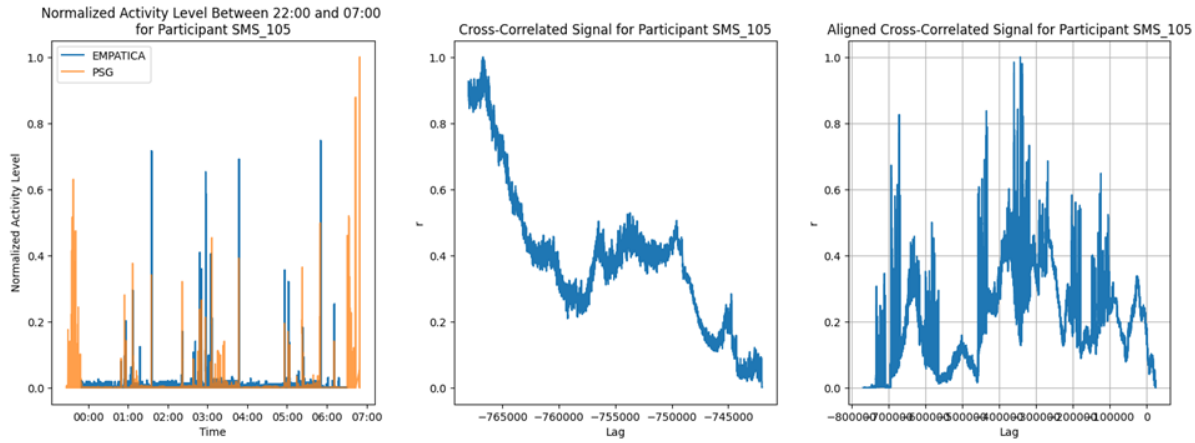
a) b) c)

Note. (a) PSG signal begins before and ends after that of EMFIT 05. (b) Correlation at lag = 0 before signal alignment, $r = 0.30$. (c) Correlation at lag = 0 after signal alignment, $r = 0.72$. This value differs from the plotted correlation at lag = 0 which peaks at $r = 1$ after realignment.

Similar challenges were encountered when cross correlating the Empatica signals with the PSG baseline. In addition to differing sampling rates and inconsistent signal recording, the lags for the cross-correlation of signals before alignment were never equal to zero. Therefore, it was not possible with the current setup to determine the correlation of the signals before alignment. After realignment, visual inspection of the plots indicated that the lag at 0 was not always the point with the highest value, suggesting incorrect realignment (Figure 5).

Figure 5

Comparison Between PSG system and Empatica for Participant 105



a) b) c)

Note. (a) PSG signal begins before and ends after that of Empatica. (b) No lag = 0 before alignment. (c) Correlation at lag = 0 after signal alignment, $r = 0.19$. A higher correlation, peaking at $r = 1$, can be observed around lag = -350,000, indicating an alignment issue.

The findings of this study are inaccurate due to unresolved issues surrounding down sampling and signal realignment exacerbated by a lack of online documentation related to these practices. Future work should focus on resolving these issues and exploring the classification of movements during sleep using machine learning methods. These techniques, combined with wearable and nearable technology, could uncover patterns undetected by traditional methods and support the diagnosis of sleep disorders in the patient's natural environment.

References

Altini, M., Altini, M., Altini, M., Altini, M., Kinnunen, H., Kinnunen, H., & Kinnunen, H. (2021). The Promise of Sleep: A Multi-Sensor Approach for Accurate Sleep Stage Detection Using the Oura Ring. *Sensors*. https://doi.org/10.3390/s21134302

Chinoy, E. D., Chinoy, E. D., Chinoy, E. D., Cuellar, J. A., Cuellar, J. A., Huwa, K. E., Huwa, K. E., Jameson, J., Jameson, J. T., Watson, C., Watson, C. H., Bessman, S. C., Bessman, S. C., Hirsch, D. A., Hirsch, D. A., Cooper, A., Cooper, A. D., Drummond, S. P. A., Drummond, S. P. A., ... Markwald, R. R. (2020). Performance of seven consumer sleep-tracking devices compared with polysomnography. *Sleep*. https://doi.org/10.1093/sleep/zsaa291

Concheiro-Moscoso, P., Groba, B., Álvarez-Estévez, D., Miranda-Duro, M. del C., Pousada, T., Nieto-Riveiro, L., Mejuto-Muiño, F. J., & Loureiro, J. P. (2023). Quality of Sleep Data Validation From the Xiaomi Mi Band 5 Against Polysomnography: Comparison Study. *Null*. https://doi.org/10.2196/42073

Eguchi, K., Nambu, M., Kamikawa, T., Ueshima, K., & Kuroda, T. (2019). Smart Textile Device with Embedded Fabric Electrodes Targeting Periodic Limb Movements Monitoring at Home: A Case Report. *Journal of Fiber Science and Technology*, *75*(11), 164–180. https://doi.org/10.2115/fiberst.2019-0020

Grandner, M. A., Grandner, M. A., Bromberg, Z., Bromberg, Z., Adams, Z., Morrell, Z., Graf, A., Graf, A., Hutchinson, S., Hutchinson, S., Hutchinson, S. A., Freckleton, D., & Freckleton, D. (2022). Performance of a Multisensor Smart Ring to Evaluate Sleep: In-Lab and Home-Based Evaluation Relative to Polysomnography and Actigraphy: Importance of Generalized Versus Personalized Scoring. *Null*. https://doi.org/10.1101/2021.12.22.21268267

Hornyak, M., Feige, B., Riemann, D., & Voderholzer, U. (2006). Periodic leg movements in sleep and periodic limb movement disorder: Prevalence, clinical significance and treatment. *Sleep Medicine Reviews*, *10*(3), 169–177. https://doi.org/10.1016/j.smrv.2005.12.003

Joseph, V., & Nagalli, S. (2024). Periodic Limb Movement Disorder. In *StatPearls*. StatPearls Publishing. http://www.ncbi.nlm.nih.gov/books/NBK560727/

Kye, S., Moon, J., Lee, T., Lee, S., Lee, K., Shin, S., & Lee, Y. S. (2017). Detecting periodic limb movements in sleep using motion sensor embedded wearable band. *2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 1087–1092. https://doi.org/10.1109/SMC.2017.8122756

Perez-Pozuelo, I., Zhai, B., Palotti, J., Mall, R., Aupetit, M., Garcia-Gomez, J. M., Taheri, S., Guan, Y., & Fernandez-Luque, L. (2020). The future of sleep health: A data-driven revolution in sleep science and medicine. *Npj Digital Medicine*, *3*(1), 1–15. https://doi.org/10.1038/s41746-020-0244-4

Periodic Limb Movement Disorder (PLMD) and Restless Legs Syndrome (RLS)—Brain, Spinal Cord, and Nerve Disorders. (2022). MSD Manual Consumer Version. https://www.msdmanuals.com/home/brain,-spinal-cord,-and-nerve-disorders/sleep-disorders/periodic-limb-movement-disorder-plmd-and-restless-legs-syndrome-rls